

Cosmic radiation has occasionally caused memory corruption in satellites and aircraft systems, forcing automatic system resets.

Single event latch-up (SEL).

Radiation can also create unintended current paths inside CMOS chips. This phenomenon is known as Single Event Latch-up (SEL). SEL may cause excessive current flow, overheating, or even permanent chip damage. Without protective circuitry, latch-up events can destroy components entirely.

Why modern chips struggle in space

Modern processors use extremely small transistor sizes such as:

- 7nm
- 5nm
- 3nm

These technologies provide higher speed and lower power consumption in smartphones and laptops. However, they are significantly more vulnerable in space environments.

Smaller transistors, smaller safety margins

Modern chips store information using extremely small electrical charges. Older processors used larger transistor structures, higher voltage margins, and physically larger memory cells. As a result, older chips are naturally more resistant to radiation effects.

Ironically, the same technological advancements that make smartphones incredibly powerful also make them more fragile in space.

Reliability matters more than speed

A Mars rover does not require gaming-level graphics or advanced multimedia capability.

Instead, it must survive radiation exposure and continue functioning for years without repair.

According to aerospace reliability studies, mission failure risks increase significantly when

untested technologies are used in harsh environments.

To survive in space, spacecraft use...

Engineers have spent decades developing ways to keep spacecraft operating in environments that would quickly overwhelm most electronics on Earth. Here are a few of the available technologies.

Radiation hardened electronics (rad-hard). These systems are specifically designed to tolerate radiation exposure.

Shielding. Shielding materials such as aluminium, titanium, and specialised composites absorb part of the incoming radiation before it reaches sensitive electronics. However, shielding increases spacecraft mass, which significantly raises launch costs.

Engineers therefore balance protection carefully against weight limitations.

Error detection and correction (EDAC). These systems in spacecraft computers continuously monitor memory and automatically correct corrupted bits. If radiation alters stored data, the system detects and restores the correct value automatically.

Triple modular redundancy (TMR). In TMR systems:

- Three processors perform the same calculation simultaneously
- Outputs are compared
- Incorrect outputs are ignored

Even if one processor experiences radiation-induced failure, the spacecraft can continue operating safely. Amazing isn't it!

RAD750 processor

Many spacecraft and Mars missions have used the RAD750 processor.

This radiation-hardened processor is based on older PowerPC architecture and operates at much lower speeds than modern consumer processors. However, it is specifically designed to survive:

RELIABILITY MATTERS MORE THAN SPEED

Consumer Electronics Priorities	Space Electronics Priorities
Performance	Reliability
Graphics capability	Stability
Compact size	Fault tolerance
	Long operational life

- Radiation exposure
- Extreme temperatures
- Long-duration missions

Its reliability makes it ideal for satellites, planetary rovers, and deep-space probes.

The future of space electronics

Although older processors remain common, modern spacecraft are gradually adopting newer technologies. Companies such as SpaceX increasingly use commercial processors, software-based fault correction, and advanced redundancy systems.

Future missions involving autonomous navigation and AI-assisted spacecraft systems will require significantly greater onboard computing capability.

However, the core philosophy of aerospace engineering is expected to remain unchanged.

Final thoughts

The use of older computers in spacecraft is not a sign of outdated engineering. Instead, it reflects the unique challenges of operating electronics in space.

Radiation, thermal stress, limited power, and the impossibility of repair make reliability the highest priority in spacecraft design.

Older processors, larger transistor structures, radiation-hardened systems, and redundant architectures provide the stability required for long-duration missions. As space technology advances, newer processors will gradually enter spacecraft systems while preserving the central engineering principle:

Reliability matters more than raw speed. **EFY**